UNIT IV

ASSIGNMENT

Class # 10. Powers and Products of Matrices:

1) Factor $A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}$ and hence compute A^{100}

ANS: Eigen values are 1, 9

$$S = [X_1, X_2] = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A^{100} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1^{100} & 0 \\ 0 & 9^{100} \end{bmatrix} \frac{1}{2} \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

2) Verify Cayley Hamilton theorem for the matrix A and find its inverse

(i) $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 3 & 2 & 4 \\ 4 & 3 & 2 \\ 2 & 4 & 3 \end{bmatrix}$

ANS: $(i)\, \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$ (ii) $\frac{1}{27} \begin{bmatrix} 1 & 10 & -8 \\ -8 & 1 & 10 \\ 10 & -8 & 1 \end{bmatrix}$

3) Using Cayley Hamilton theorem find A^8, if $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$

ANS: $625I$